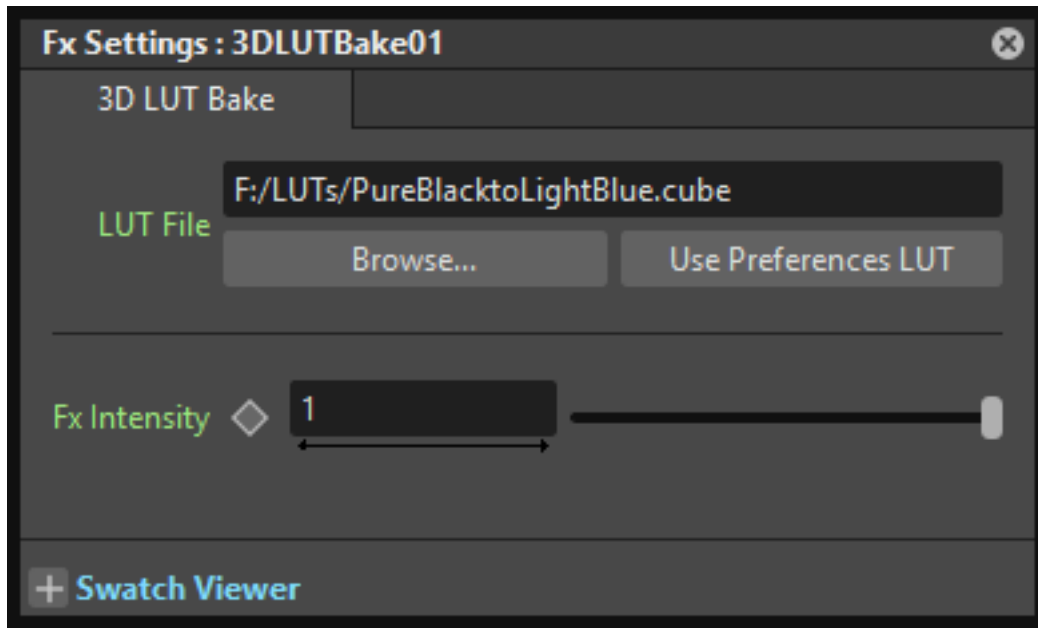


3D LUT Bake FX

OpenToonz | Image Adjust | User guide



Apply a selected 3D LUT to rendered image colors. Connect the image or composition to Source. For a final overall look, place this node after the completed composition. Alpha is preserved.

Controls and accepted entries

LUT File - Enter the path to an existing, readable **.cube** or **.3dl** file. Use a full absolute path, for example F:/LUTs/PureBlacktoLightBlue.cube. Do not enter a web URL or surround the path with quotation marks. A blank field passes the input through. The path is not animated; the LUT file itself is not embedded in the scene.

Browse... - Opens a file picker filtered to .cube and .3dl files. Select a supported LUT; canceling leaves the current value unchanged. The selected file is validated before its absolute path is assigned.

Use Preferences LUT - Copies the LUT path configured for the current monitor in Preferences into this FX. It can copy a configured path even when display calibration is disabled. It validates the file before assigning it. Later Preferences or monitor changes do not change this stored path.

Fx Intensity - Enter a number from **0 to 1**, or use the slider. The default is **1** (full effect); **0** gives no contribution from the effect; intermediate values blend its contribution. The diamond is the standard keyframe control for animating intensity. Intensity is not a substitute for fixing an invalid LUT path.

Swatch Viewer (+) - Expands the FX settings preview area. It is a viewing control, not a LUT parameter, and accepts no text or numeric entry.

Supported files and viewing results

Accepted LUT data

.cube: A 3D table with LUT_3D_SIZE from **2 to 129**, followed by exactly size cubed RGB triples containing finite floating-point numbers. TITLE and comment lines are accepted. DOMAIN_MIN / DOMAIN_MAX may specify three input bounds, or LUT_3D_INPUT_RANGE may specify one shared minimum and maximum; do not combine these two forms. Each minimum must be below its maximum. Without domain tags, the input range is 0 to 1. 1D, 2D and shaper LUTs are not supported.

.3dl: The supported variant begins with 3DMESH, then Mesh [input bit depth] [output bit depth], an input-grid line, and integer RGB triples. Input bit depth is 0 to 7; output bit depth is 1 to 30. The grid size is 2 raised to the input bit depth, plus 1. Other .3dl dialects may not load; the extension alone does not establish compatibility.

Display calibration and baked output

This FX changes the rendered image data. Preferences display calibration changes how the image is shown on the current monitor. A monitor-calibration LUT normally compensates for that particular display; baking it does not guarantee the same appearance on another monitor.

If the same LUT is applied by both this FX and display calibration, the displayed baked result receives the transform again. When comparing deliberate baking of that same transform, account for this extra display application. Do not generally disable monitor calibration when evaluating a separate creative look LUT.

The FX evaluates LUT values in nonlinear RGB; the FX framework handles surrounding linear-color conversions. Baking does not inherently remove an image from the linear-rendering workflow. A reported Preview/output mismatch is still under investigation: compare equivalent render settings and distinguish saved pixel values from display appearance.

Limits and troubleshooting

LUT inputs are clamped to the declared domain and the current FX clips transformed RGB to 0-1, including floating-point output. It is not an unrestricted HDR transform. Transparent edges are processed with alpha taken into account.

A missing or malformed selected file causes an error; it is not automatically replaced with an identity LUT. Browse and Preferences-copy validation show a warning; manually entered invalid paths can fail when the FX renders. Move or repair the file and update the path as needed. Keep external LUT files available when transferring a scene to another computer.

Help and version

Use the Help icon in FX Settings to open this guide. This guide describes the current controls shown above, including the button label Use Preferences LUT. Planned reusable file-field and error-reporting changes are not presented as implemented behavior.